

Unit 9, Lesson 3: Cross Sections



Benchmark

MA.912.GR.4.1 Identify the shapes of two-dimensional cross-sections of three-dimensional figures.



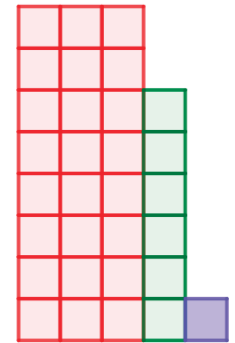
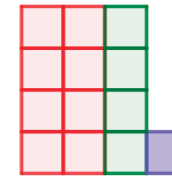
Learning Target

- ☐ I can identify the shapes of two-dimensional cross sections of three-dimensional figures.



9.3.1 Warm-Up: Patterns

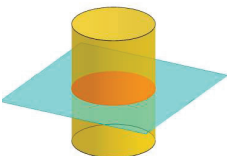
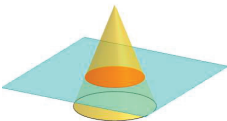
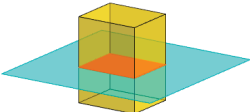
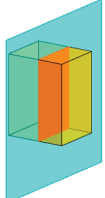
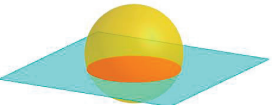
- How many red squares are in the seventh pattern?





9.3.2 Collaboration: Cross Sections

The table below shows the resulting cross section for each type of slice.

Figure	Cross Section	
	Horizontal Plane	Vertical Plane
Cylinder		
Cone		
Rectangular Prism		
Triangular Prism		
Sphere		

1. What do you notice about the diagrams for horizontal slices?
2. What do you notice about the diagrams for vertical slices?
3. Discuss with your partner whether or not the shape of a cross section will change if the height of the plane is changed. Summarize your discussion.
4. Complete the table by describing the shape of the cross section that is created when the figure is sliced by each.

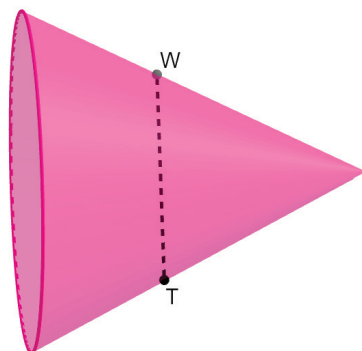
Figure	Cross Section Shape	
	Horizontal Plane	Vertical Plane
Cylinder		
Cone		
Rectangular Prism		
Triangular Prism		
Sphere		



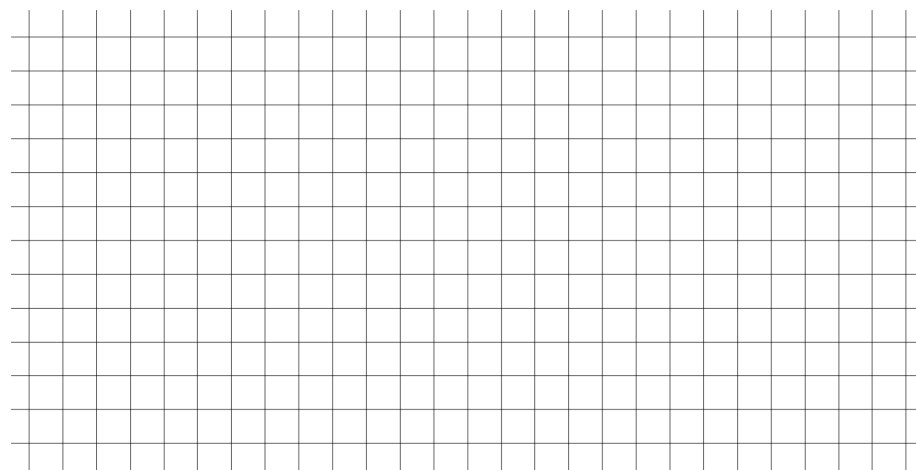
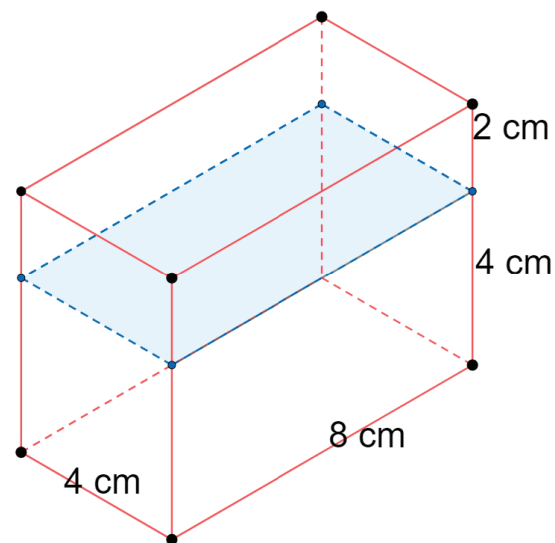
9.3.3 Guided Instruction: Drawing Cross Sections

Three-dimensional figures can be oriented in different ways. Cross sections are typically described by their relation to the base of the three-dimensional figure. For example, cross sections are often described as parallel or perpendicular to the base of the figure.

1. A cone is shown with a cross section parallel to the base. Draw the cross section of the cone on the grid, given $WT = 6$ units.



2. A rectangular prism is shown with a cross section parallel to the base. The dimensions are given in centimeters (cm). Draw the cross section on the grid and label its dimensions.



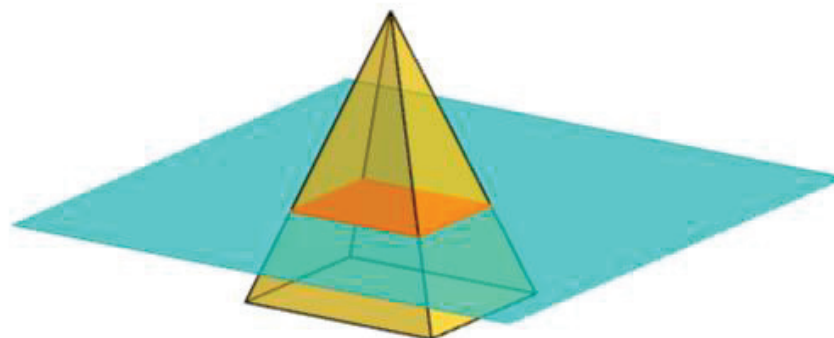
3. A sphere with radius of 4 inches (in.) with a horizontal cross section is shown.



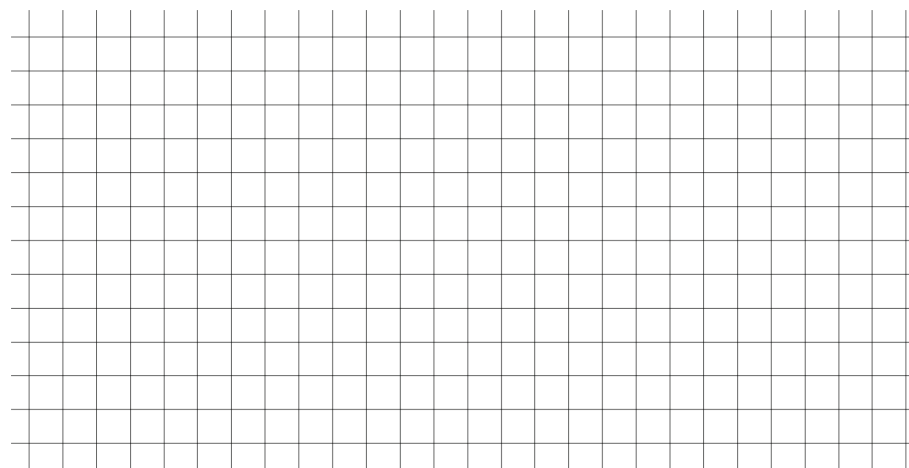
Draw the cross section if it occurs $\frac{3}{8}$ of the distance from the sphere's center.



4. A rectangular pyramid is shown, where the dimensions of the base are 12 centimeters (cm) by 21 cm and the pyramid's height is 33 cm.



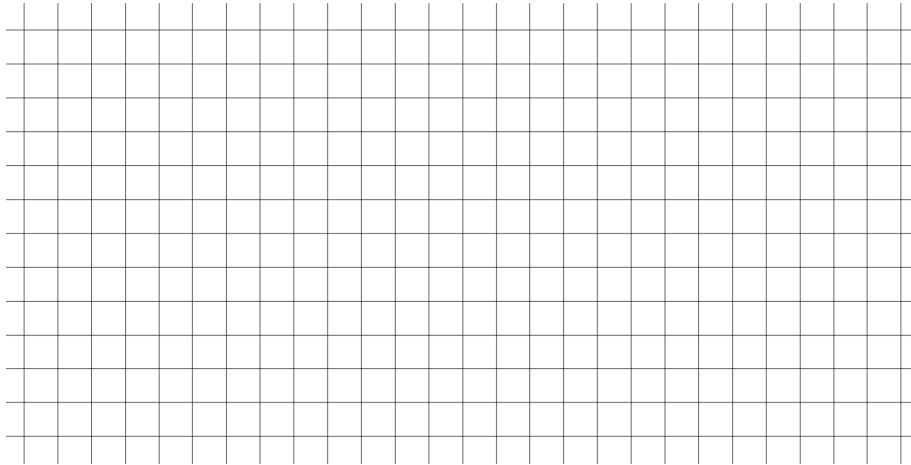
Draw the cross section parallel to the base that is located at a height of 11 cm.





9.3.4 Your Turn!

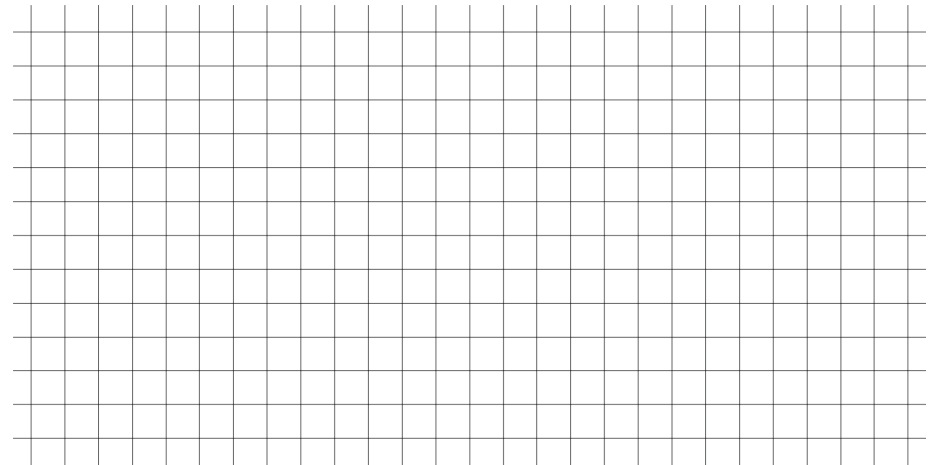
1. Draw the cross section formed by the intersection of a sphere with a diameter of 15 millimeters and a perpendicular plane passing through the center of the sphere.



2. A cylinder has a radius of 4 inches (in.) and a height of 15 in. Draw the cross section formed by the intersection of a cylinder and a plane perpendicular to the base passing through the center of the cylinder.



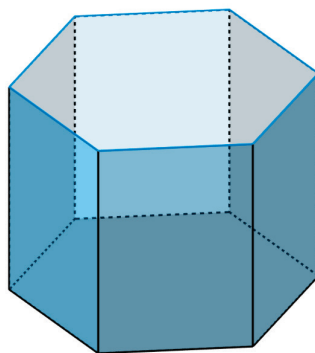
3. Draw the cross section formed by the intersection of a pentagonal prism and a plane parallel to the base passing through the prism.





9.3.5 Wrap-Up: Ticket Out the Door

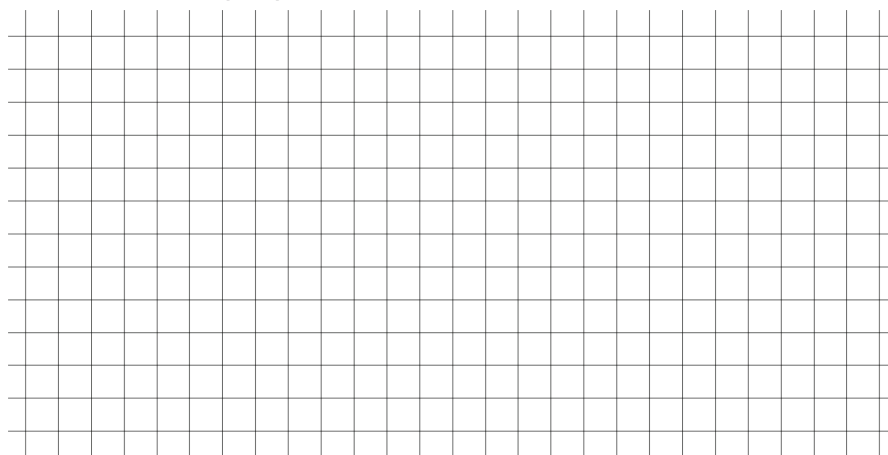
1. A hexagonal prism is shown.



- a. Draw the cross section that results from a plane that is parallel to the base.



- b. Draw the cross section that results from a plane that is perpendicular to the base.



Unit 9, Lesson 4: Solids of Rotation



Benchmark

MA.912.GR.4.2 Identify three-dimensional objects generated by rotations of two-dimensional figures.



Learning Target

- ☐ I can identify three-dimensional objects generated by rotations of two-dimensional figures.